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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/096,681	11/12/2020	Michael W. Olszowy	1160015USP	8384

16275 7590 07/09/2024

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EXAMINER
KINSEY WHITE, NICOLE ERIN

ART UNIT	PAPER NUMBER
1648	

NOTIFICATION DATE	DELIVERY MODE
07/09/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docket@hpdlaw.com
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Office Action Summary	Application No. 17/096,681	Applicant(s) Olszowy, Michael W.	
	Examiner Nicole Kinsey White	Art Unit 1648	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 17 April 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-2,4-11,13-15 and 19-28 is/are pending in the application.

5a) Of the above claim(s) 5 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2,4,6-11,13-15 and 19-28 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Withdrawn Rejections

The rejection of claims 1-2, 7-11, 13-14 and 19-27 under 35 U.S.C. 103 as being unpatentable over Leisi et al. (Journal of Membrane Science, March 5, 2020, 603:118012, pages 1-14), Kayukawa et al. (U.S. Patent Application No. 2023/0103302; effectively filed February 28, 2020) and Rockey et al. (Current Opinion in Biotechnology 2019, 57:42–49) has been withdrawn in view of applicant's amendments to claim 1.

The rejection of claims 4 and 6 under 35 U.S.C. 103 as being unpatentable over Leisi et al. (Journal of Membrane Science, March 5, 2020, 603:118012, pages 1-14) and Strauss et al. (Biotechnology and Bioengineering, 2009, 104(2):371-380) has been withdrawn in view of applicant's amendments to claim 1.

The rejection of claim 22 under 35 U.S.C. 103 as being unpatentable over Leisi et al. (Journal of Membrane Science, March 5, 2020, 603:118012, pages 1-14) and AAT Bioquest (Safety Data Sheet, Revision Date March 27, 2023) has been withdrawn in view of applicant's amendments to claim 1.

Depending on how applicant resolves the rejections under 112 below, these rejections may be reinstated.

Claim Objections

Claim 1 is objected to because of the following informalities: For clarity, claim 1 should be amended to recite "subjecting an evaluation solution to a purification process~~ing~~ to ~~prepare~~ create a treated solution, wherein the evaluation solution comprises fluorescent viral particles, and wherein the purification process~~ing~~ comprises at least one viral removal technique for separation of a viral contaminant" and to recite "wherein the determining a degree of removal of the fluorescent viral particles comprises subjecting a sample of the treated solution to ~~an analytical technique~~ flow cytometry to analytically quantify a residual concentration of the fluorescent viral particles in the treated solution ~~from fluorescent properties of the fluorescent viral particles, and the analytical technique comprises flow cytometry~~".

Claim 8 should recite "which second said fluorescent emission signature may be the same or different than the first said fluorescent emission signature".

Appropriate correction is required.

NOTE

Applicant is advised to review each claim for clarity. For example, much of claim 1 is confusing and should be amended to be succinct and clear.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

> The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1-2, 4, 6-11, 13-15 and 19-28 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre- AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claims 1, 11 and 26 recite the phrase "wherein the flow cytometry comprises detecting for a fluorescent emission signature of the fluorescent dye of the fluorescent viral particles and the method is in the absence of fluorescent staining of the sample of the treated solution prior to the flow cytometry and in the absence of fluorescent staining the treated solution prior to obtaining the sample of the treated solution". The wherein clause is confusing and should be rewritten. Further, it is not clear what is meant by "the method is in the absence of fluorescent staining of the sample of the treated solution prior to the flow cytometry and in the absence of fluorescent staining the treated solution prior to obtaining the sample of the treated solution". The phrase requires that the method (assuming the method of claim 1) is in the absence of fluorescent staining of the sample prior to flow cytometry and prior to obtaining the sample of the treated solution. The first step of claims 1, 11 and 26 use an evaluation solution that has fluorescent viral particles in it. Thus, it is not clear how the method is "in the absence of fluorescent staining" when fluorescent particles are used in the first step. Accordingly, one of ordinary skill in the art will not know the metes and bounds of the claim.

Claim 15 is confusing. It is not clear what is meant by the phrase "using the feed volume of the preliminary feed solution and the reagent volume of the viral standard

solution combined to make the evaluation solution and the analytically quantified number concentration of the viral standard solution". Further, it is not clear what type of calculation is being performed. Accordingly, one of ordinary skill in the art will not know the metes and bounds of the claim. The claim should be rewritten to clarify what is being claimed.

This rejection also affects claims depending from claims 1, 11, 15 and 26.

Conclusion

No claim is allowed.

Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Nicole Kinsey White whose telephone number is (571)272-9943. The examiner can normally be reached M to Th 6:30 am to 6:00 pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Janet Andres can be reached on 571-272-0867. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/NICOLE KINSEY WHITE/
Primary Examiner, Art Unit 1648